

Application: Crops in silico 1.0

Amy Marshall-Colon - amymc@illinois.edu
Direct Fund Projects

Final Scientific Progress Report

Completed - Mar 29 2019

Form for "FFAR Annual Progress Report"

Grant Information*

Award Program	Direct Fund Projects
Grant ID	0000000012
Project Title	Crops in Silico 1.0
Grant Start Date	06/16/2017
Grant End Date	12/31/2018
Total Budget	601126.46
Matching Funder(s)	iSEE and NCSA
Final Report Date	03/31/2019

Project Director/Principal Investigator Information*

Full Name	Dr. Amy Marshall-Colon
Email	amymc@illinois.edu
Phone	217-300-1430

Grantee Organization Information*

Organization Name	The Board of Trustees of the University of Illinois
Address	1901 South First St. Ste A
City	Champaign
State	Illinois
ZIP	61820
EIN	37-6000511

Authorized Signing Official Information*

Name	Avijit Ghosh
Email	spa@illinois.edu
Phone	217-333-2187

1. GENERAL INFORMATION

1.1. Please list the geographic location(s) - city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.*

University of Illinois - Urbana Champaign, Urbana, IL, IL-13 congressional district
Pennsylvania State University, State College, PA, PA-12 congressional district

1.2. How many new jobs were created by the grant during this reporting period? *

3

1.3. How many jobs were maintained by the grant during this reporting period? *

12

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant?*

No

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization. *

No

2. SCIENTIFIC REPORT

2.1. Public Abstract*

The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Current models predict an increasing gap between food supply and demand over the next 50 years. There is uncertainty that US breeding programs will be able to develop cultivars within this time frame to mitigate the yield discrepancy, especially at a time when land area and water resources are limiting to agricultural productivity. There is an immediate need for technology capable of predicting the fitness of various crop phenotypes in response to climate and resource availability. The aim of the Crops in silico 1.0 project was to lay the foundation for generating virtual plant models that accurately capture whole system dynamics in response to in silico environmental and genetic perturbations. A goal of this research project was to identify a feasible approach to link models across biological scales, specifically from genes to phenotypes. This multiscale modeling would enable fine and coarse analyses to observe whole system response at the different scales, and predict ideotypes for different objectives and environments. To achieve this goal, we built a computational framework that allows asynchronous communication between models without altering their original code or timesteps. This framework allows the exchange of inputs and outputs between models, mimicking the flow of biological information across scales. We used the framework to successfully build a multiscale model of the soybean photosynthetic response to elevated CO₂ concentration [CO₂]. The vetted model was used to perform in silico genetic perturbations and simulate the anticipated outcomes on photosynthetic efficiency. After dozens of simulations, we identified three top transcription factor gene candidates that will be tested experimentally for their regulatory role in controlling photosynthetic efficiency. The computational framework was also used to link whole plant models of roots and shoots to form the core structure of virtual soybean and corn plants. These linked models were subjected to advanced visualization approaches so that in silico observations can be made on an entire stand of plants. The use of 3-dimensional architecture was shown to improve model simulations of light capture throughout a soybean canopy, providing new insights about plant competition for light resources.

2.2 Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making. *

...The Crops in silico framework has seen considerable interest from multiple domains as well as within the fields of plant biology and crop sciences. Following its presentation at SciPy 2018, an interdisciplinary scientific software conference, there have been initial collaboration efforts from researchers in several different disciplines (such as health sciences). In conjunction with the model repository and the user interface for developing integrations, we believe that the interface is already making an impact on the scientific and industrial communities. The interface is open and available for anyone to use and is equipped with complete documentation and tutorials. Thus, this tool is immediately available to the agriculture community and is likely of interest to ag industry.

The scientific findings of our work are not immediately applicable to US agriculture, because after only one year we are still in the discovery phase. In the next iteration of our work, we will take model predictions and test them in plants under controlled environments. Transgenic plants that show a promising increase in photosynthetic efficiency in growth chambers and greenhouses will be tested in the field at the Soy-FACE facility. Positive results at this stage will be poised to translated our findings into agricultural crops that may then have a direct impact on food production in the US.

2.3. Research Outcomes/Impact*

The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- Potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- Intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- End outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes

We believe this project will have a substantial impact across fields, and that the collaborations forged through this project will provide a template for other interdisciplinary projects. The Crops in silico project outcomes will have a direct impact on the food and agriculture community, and food security, through the design and development of ideotypes based on model simulations and predictions. The Crops in silico platform will also impact crop science research by allowing researchers to ask questions in which they can perform simulations to arrive at new hypotheses and predictions. The advanced visualization and user interface will make the in silico experiments accessible to a broad community including educators and students. Cinema quality simulations can also be used to educate policymakers in a tangible way.

Intermediate outcomes

The Crops in silico user interface is immediately available for experimentalists, modelers, and educators to use through GitHub where they will also find documentation and tutorials. We have made all code, models and methods available to the public. Our current model predictions are also publically available on BioRxiv and have been submitted to the journal in silico Plants, which is also open access.

End outcomes

Ultimately, the Crops in silico project will provide a useful tool for the scientific and agriculture communities. The model plants will be able to provide reliable predictions for users to explore how crops will respond to genetic and environmental perturbations. This will have a direct impact on food security since this tool can be used as the first step in the pipeline to generate ideotypes that will perform well in challenging growing conditions.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. *

Specific Aims:

1. Create whole virtual crops by integrating detailed above and below ground models.
2. Subject the virtual crops to in silico perturbations to test prediction accuracy of individual vs integrated models against experimental data, toward the development of crop ideotypes.
3. Construct the Crops in silico infrastructure that will include software for the orchestration system, conversion of source code to executables, and a user-friendly interface.

4. Create advanced, 3-D visualization of the integrated model outputs.

Milestones (date completed)

Model integration - Individual models will be integrated using our proposed orchestration system and optimized. (03/30/18)

Perform quality assurance on integrated models - Integrated model results will be analyzed and validated using existing data from field experiments. This milestone will provide information on the prediction accuracy of the integrated bean and maize models and identify genetic targets for future engineering and experimental efforts. (06/04/18)

Build a user interface - The collaborative funds from NCSA will be used to build an interface capable of cataloging models and incorporation of software and functions for authorization and authentication services, job execution and status tracking, and results tracking. (12/01/18)

Model communication - Provide tutorials and worked examples for adding model communication libraries to existing models and explore alternative methodologies. (03/30/18)

Model transformation - Will use a Dockerfile recipe to build Cis-compliant models by hand; enables users to upload executable files rather than source code to encourage participation by the community. (12/15/18)

Disseminate results - finalize model documentation and prepare manuscripts and presentations for annual Cis meeting. (12/15/18)

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award?*

No

2.6. What was accomplished under the goals/specific aims or milestones for the project? *

Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include a discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like.

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Researchers were able to meet all stated goals.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.*

In our proof of concept multiscale modeling, we rigorously tested each individual model using standardized protocols in the field, including sensitivity analysis and parameter sweeps on the models. The integrated model outputs were vetted against field-observed data to ensure biological accuracy. All new model predictions are in an experimental pipeline to test gene function and regulatory impact on plant phenotype. The manuscript that describes the results of the proof of concept model is currently under rigorous peer review at in silico Plants journal, and is also open for public comment on BioRxiv.

2.8. List key stakeholders that could be served by results of this project.*

The direct stakeholders in this project range from our colleagues in plant biology, to educators and students, to funding agencies and private companies developing new methodologies to improve crop yield, and ultimately to the consumers who will in the future rely on this increased yield.

2.9. How have the results of this reporting period been disseminated to communities of interest? *

Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct

similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products.

Crops in silico Annual Symposium, Workshop, and Hackathon: The annual symposium/workshop is an opportunity to bring together a diverse community of plant modelers, experimentalists, and computer scientists to discuss the goals and objectives of Crops in silico. The Illinois team uses this opportunity to disseminate results of the project.

The Second Annual Crops in silico Symposium and Workshop was June 26-28, 2017 in Oxford. The meeting covered recent developments in plant modelling and model integration, and users and usability of models. The goal of the workshop was to foster new research collaborations.

The Third Annual Crops in silico Symposium and Workshop was July 31-August 3, 2018 at NCSA. The objective of the Symposium and Workshop was to identify how the Cis community can communicate and coordinate their research, training and educational activities. New this year was a programming workshop and a Hackathon. At the workshop, trained Software Carpentry instructors taught researchers basic computing skills to enhance their Hackathon experience. The Hackathon was run by Matt Turk and Meagan Lang and consisted of short tutorials alternating with practical exercises that guided participants through model integration using Cis_interface that enables model communication, orchestration, and transformation. The goals of the Hackathon were to improve the framework through collaboration and build social capital (a Cis user base). Both the symposium/workshop and hackathon were well attended, drawing attention to the FFAR-funded Crops in silico project.

Scientific presentations: A number of scientific presentations about the Crops in silico project have been made during the funding period. For the most part, these have been invited presentations by conference organizers.

- Yu Wang. Multiscale modeling and visualization platform for crops. 3rd Annual Crops in silico Symposium and Workshop. July 31 - Aug 3, 2018. Urbana, IL.
- Amy Marshall-Colon. Integrative Modeling and Visualization for the Development of in Silico Crops. Plant Biology 2018. July 14 - 18, 2018. Montreal, Canada.
- Meagan Lang. Cis_interface: A Python package for connecting scientific models across scales and languages. SciPy. July 9 - 15, 2018. Austin, TX.
- AJ Christensen. Advanced Visualization for Crops in silico. School of Information Sciences, University of Illinois. March 12, 2018, Illinois.
- Balaji Selvam and Diwakar Shukla. Revealing the conformational flexibility of dual affinity nitrate

transporter (NRT1.1) gene in Arabidopsis Thaliana. Monsanto Research Symposium. June 30, 2017, Urbana, Illinois. Won best poster award.

- Meagan Lang. Developing an Integration Framework. 2nd Crops in silico Symposium and Workshop. Jun 27, 2017, Oxford England.

- Balaji Selvam, Shriyaa Mittal and Diwakar Shukla. Understanding the conformational switches in membrane transporter proteins using Blue Waters. Blue Waters Symposium, May 16-19, 2017, Oregon.

Public engagement:

Appearance on the WCIA Morning Show to discuss the Crops in silico project, aired 03/26/19, <https://www.wcia.com/the-morning-show/u-of-i-grant-studying-plants-virtually/1878491488>

Participant in the Genomics for Journalists event at the IGB, 03/25/2019.

News Gazette featured the FFAR award for Crops in silico, "UI earns \$5 million federal grant to expand project combining ag, tech," published 03/16/19 (top of front page), <http://www.news-gazette.com/news/local/2019-03-16/ui-earns-5-million-federal-grant-expand-project-combining-ag-tech.html>

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them.*

Only describe significant challenges that may have impeded the research and emphasize their resolution.

There was difficulty in combining the SimRoot and BioCro models due to extreme incompatibility of model structures. Team members at UIUC and Penn State worked through this issue to combine the models in a way that the integrated model simulations were able to recapitulate what has been observed in nature for carbon assimilation in corn. Although the solution worked, it is not sustainable for long-term model integration. Combined root and shoot models will be the core for the virtual crop models, so a more sustainable solution is in place for the next iteration of the project. The plan is to re-write parts of the SimRoot model so that it is universally compatible with other types of models.

Additional difficulties arising from incompatibilities within model integrations were anticipated -- for instance timestepping and unit conversion -- and explicitly planned for in the development of the integration framework.

2.11. What opportunities for training and professional development has the project provided during this reporting period?*

If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals.

In-depth training in participants’ discipline occurred in their own research labs. These included technical skills, experimental design, and data analysis and interpretation.

IDPs focused on developing professional and interdisciplinary research for biologists and computer scientists. This was accomplished during team meetings, cross-disciplinary training and conferences.

Weekly interdisciplinary team meetings allowed participants to develop professional skills, including communication skills to understand the languages of other disciplines, research methods and cultures of different disciplines, and how to target different audiences. Team meetings altered between discussion and working meetings. The goal of research discussions (e.g., research updates, discussion of roadblocks and solutions, sharing experiences and lessons learned) was to improve students’ ability to design interdisciplinary research projects and give them a better understanding of the related issues and challenges. Working meetings provided time for collaboration and troubleshooting. Postdocs developed leadership skills by organizing and running the weekly team meetings.

Participants received training in the interconnected disciplines. Biologists attended Software Carpentry workshops which teach basic research computing skills and cover basic concepts and tools, including program design, version control, data management, and task automation. Computer scientists assisted in plant architecture data collection in the field and greenhouse.

Participants built interdisciplinary networks and communicated their work with formal poster and oral presentations at cross-disciplinary conferences and symposia (eg Crops in silico Symposium and Workshop).

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail.*

- 1 researcher (Justin Mcgrath)
- 3 postdocs (Yu Wang, Ghana Challa, Balaji Panneerselvam)
- 3 grad students (Stuti Shrivastava, Kavya Kannan, Xiyu Yang)

3. INFORMATION PRODUCTS

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.*

Scientific presentations: A number of scientific presentations about the Crops in silico project have been made during the funding period. For the most part, these have been invited presentations by conference organizers.

- Yu Wang. Multiscale modeling and visualization platform for crops. 3rd Annual Crops in silico Symposium and Workshop. July 31 - Aug 3, 2018. Urbana, IL.
- Amy Marshall-Colon. Integrative Modeling and Visualization for the Development of in Silico Crops. Plant Biology 2018. July 14 - 18, 2018. Montreal, Canada.
- Meagan Lang. Cis_interface: A Python package for connecting scientific models across scales and languages. SciPy. July 9 - 15, 2018. Austin, TX.
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- Balaji Selvam, Shriyaa Mittal and Diwakar Shukla. Understanding the conformational switches in membrane transporter proteins using Blue Waters. Blue Waters Symposium, May 16-19, 2017, Oregon.

3.2. Please provide a list of citations for the information products produced during this period. *

Cis_interface software: https://github.com/cropsinsilico/cis_interface

CiS Github Organization with model repos: <https://github.com/cropsinsilico>

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award?*

Yes

If yes, please provide citation.

Challa, G, Marshall-Colon, A†. 2019. in silico crops and multi-omic approaches to meet agricultural challenges. CAB Reviews, Accepted: Ms. No. PAVSNNR-D-18-00354R1

Christensen, A. J., Srinivasan, V., Hart, J. C., & Marshall-Colon, A. (2018). Use of computational modeling combined with advanced visualization to develop strategies for the design of crop ideotypes to address food security. *Nutrition reviews*, 76(5), 332-347.

<https://academic.oup.com/nutritionreviews/article/76/5/332/4942300?questAccessKey=64fa920e-75c9-47eb-b2ab-dd3327070d38>

Kannan, K, Wang, Y, Challa, GS, Lang, MM, Srinivasan, V, Long, SP, Marshall-Colon, A†. Combining gene network, metabolic, and leaf-level models show means to future-proof soybean photosynthesis under rising CO₂. *bioRxiv*: doi.org/10.1101/582981

Lang M. 2019. yggdrasil: A Python Package for Integrating Computational Models Across Languages and Scales. *in silico Plants*.

Shrivastava, S., Marshall-Colon, A.† 2019. Big Data in Agriculture and Their Analyses. In: Ferranti, P., Berry, E.M., Anderson, J.R. (Eds.), *Encyclopedia of Food Security and Sustainability*, vol. 1, pp. 233-237. Elsevier. ISBN: 9780128126875

Conference abstracts:

Marshall-Colon, A. 2018. Plant Biology 2018. Integrative Modeling and Visualization for the Development of in Silico Crops.

American Geophysical Union (AGU) Fall Meeting, Friday, 16 December 2016. San Francisco, CA. Crops in silico: A community wide multi-scale computational modeling framework of plant canopies. Venkatraman Srinivasan, Andrew Christensen, Kalina Borkiewicz, Xu Yiwen, Apollo Ellis, Balaji Panneerselvam, Kavya Kannan, Stuti Shrivastava, Donna Cox, John Hart, Amy Marshall-Colon, and Stephen Long

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.*

Crops in silico website: <https://cropsinsilico.org>

Documentation and tutorials: https://cropsinsilico.github.io/cis_interface/

Mockup of interface: <http://plants.hub.yt/LeM>

YouTube video: Connecting Scientific Models across Scales & Languages w/ Python | SciPy 2018.
<https://www.youtube.com/watch?v=Vv6Ksqitfc8>

3.5. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period?*

No

3.6. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements? *

No

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?*

All manuscripts are published in open access journals including BioRxiv so that the public has early access to our results and approaches. All models and code are available as open access and open source through GitHub with information on the model input/outputs and execution displayed on the landing page of the repository. Documentation is also provided on using the produced framework to integrating models, including a walkthrough from installation to execution that was utilized during the 2018 Crops in Silico Workshop and Hackathon.

4. DATA MANAGEMENT

4.1. During this period, did the project generate any data?*

Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries).

No

Please list the data generated in this reporting period.

N/A

4.2. If you list multiple data sets, are these data sets related?

No

4.3. Please provide copies of relevant metadata records to support FFAR’s mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.

FFAR is developing a more nuanced Data Management Policy. As part of this effort, we are introducing new questions into the Annual Reports on Data. All questions on this page are required to have a response. In cases where your answer may be repeated, you are welcome to copy-paste relevant information. Thank you for your patience as we update our reporting.

Are the data experimental, observational, simulation, or compiled/derived? Check all categories that apply.

No Responses Selected

For each checked box, please provide a brief description of the data including where the data are collected / sourced.

(No response)

Describe (what, where, when) and list additional data (not captured in question above), plus samples, physical collections, software, curriculum materials and other materials produced during this period. Who is the expert or person(s) tasked with data collection (please provide their name and affiliation with the project)? This can be any member of the team. For data-focused projects, that individual is expected to be key personnel.

(No response)

Provide how the data are organized and managed for all data created and utilized during this project. This could include the relevant data standardization, data formats, data definitions (e.g., data dictionary, ontologies), data organization, and/or where the data is stored. Please note if any of these data are confidential, and if so, when you would anticipate being able to share them publicly in the future.

(No response)

FFAR supports the FAIR principles. We would like to understand how your project aligns with FAIR principles (for definition, see [Go Fair](#), for more information see [Nature Magazine](#)). Does your data align with the 4 principles in FAIR? Please check all boxes that apply and provide a brief description:

No Responses Selected

For each checked box, please provide a brief description.

(No response)

What are the provisions for the data for the appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements?

(No response)

Describe the plans for archiving data, samples, and other research products, and for preservation of access to them. Refer to the [FFAR Data Guidance Document](#) for more information

(No response)

Will this project coordinate with existing FFAR-supported grants and their data products? If so, provide a brief description of how, including the project title and PI of the existing FFAR work. If not, provide a brief rationale for why this project does not coordinate with existing FFAR-supported work.

(No response)

Please provide any feedback on these new data questions including any areas where you would like to see more clarity or explanation.

(No response)

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked 'owner' in parentheses on the menu on the left of the screen when you open your award) can hit 'Submit' on finished tasks. If you are not the owner, the 'Submit' button will not appear. FFAR does not receive notification of completed tasks without a submission.

5. CERTIFICATION

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information.

By signing below, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name*

Dr. Amy Marshall-Colon

PI Electronic Signature*

A handwritten signature in black ink on a light gray background. The signature is cursive and reads "Amy McLean".

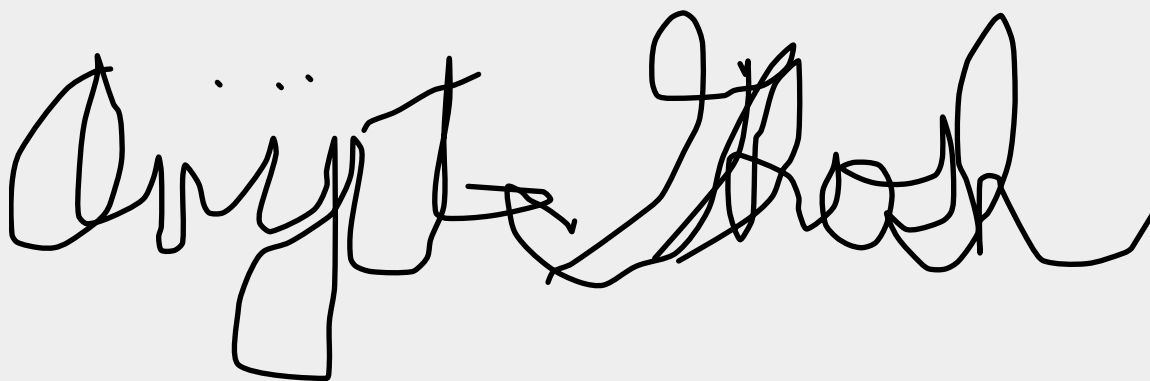
Date*

03/29/2019

Authorized Signing Official (ASO)*

Dr. Avijit Ghosh

ASO Electronic Signature*

A handwritten signature in black ink on a light gray background. The signature is cursive and reads "Avijit Ghosh".

Date*

03/29/2019

Final Invention Report and Certification

Completed - Mar 29 2019

Form for "Final Invention Report and Certification"

FINAL INVENTION REPORT AND CERTIFICATION

A Final Invention Report must be submitted within 90 days after the expiration or termination of a FFAR funded research grant. The Report must include all inventions which were conceived or first actually reduced to practice during work performed under the funded grant, from the original effective start date of the grant through the date of completion or termination. The Final Invention Report and Certification in no way relieve the principal investigator, project director, or the grantee, of the obligation to assure that all inventions are promptly and fully reported directly to the Foundation for Food and Agriculture Research (FFAR), as required by terms of the grant or award.

If no inventions were conceived or reduced to practice during the funded research, write a statement to that effect in the first block under item Title of Invention. The Authorized Organization Representative (AOR) must certify the Report prior to submitting it.

Grantee:*

The Board of Trustees of the University of Illinois

Grant ID:*

515760

Project Title:*

Crops in Silico 1.0

Date of the Report:*

03/28/2019

INVENTION(S)

If no inventions were conceived or reduced to practice during the funded research, write "NONE" in the space provided below.

	Name of Inventor	Title of Invention	Date Reported to FFAR
1	None		
2			
3			
4			
5			
6			
7			
8			
9			
10			

The undersigned hereby certifies that to the best of their knowledge and belief, the information in the Report below are true, accurate, and complete. You also certify that the Report lists all inventions which were conceived and/or first actually reduced to practice during the work under the above-referenced FFAR grant or award for the period of:

Project Start Date:*

06/17/2017

Project End Date:*

12/31/2018

Authorized Organization Representative (AOR) Name:*

Avijit Ghosh

AOR Title:*

Comptroller

AOR Signature:*

A handwritten signature in black ink on a light gray background. The signature is written in a cursive, flowing style. The first part of the signature, 'Avijit', is written with a large, looped 'A' and a series of connected loops for the 'v', 'i', 'j', 'i', and 't'. The second part, 'Ghosh', starts with a large 'G' and continues with a series of loops for the 'h', 'o', 's', and 'h'. The signature is centered horizontally within the gray box.

Date:*

03/29/2019

PI Signature:*

A handwritten signature in black ink on a light gray background. The signature is stylized and appears to be 'Anthony M. G. R.'.

Date:*

03/28/2019

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked 'owner' in parentheses on the menu on the left of the screen when you open your award) can hit 'Submit' on finished tasks. If you are not the owner, the 'Submit' button will not appear. FFAR does not receive notification of completed tasks without a submission.

Final Financial Status Report

Completed - Mar 29 2019

Please download the [Final Financial Status Report](#) and upload the completed form.

[C2522 FFAR Final FSR](#)

Filename: C2522_FFAR_Final_FSR.xlsx **Size:** 29.4 kB

Final Equipment Inventory Report

Completed - Mar 29 2019

[Equipment](#)

Filename: Equipment.pdf **Size:** 28.4 kB

Copies of Relevant Metadata Records

Completed - Mar 29 2019

Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products.

[metadata report](#)

Filename: metadata_report.pdf **Size:** 28.0 kB

Other Modifications

Incomplete

No Cost Extension Request

Incomplete

Form for "No Cost Extension Request"

No Cost Extension Request Instructions

A No Cost Extension is required when a grantee determines that their activities will not be completed during the approved project timeline. A no-cost extension requires prior approval from the Director of Grants Management. FFAR grants and cooperative agreements are awarded under a project period system. A project may be approved for multiple years (usually 3-5 years), and is generally funded in annual increments known as budget periods. With rare exceptions, budget periods are 12 months in duration. FFAR expects that grantees will complete all requirements of an award by the project period end date; however, a one-time no cost extension may be requested if the action is not prohibited in the grant agreement. The Director of Grants Management can approve up to 6 additional months but no additional funds are awarded to complete the funded tasks. Note that a No Cost Extension Request should not be submitted for the sole purpose of expending remaining funds – such request will be disapproved. No Cost Extensions cannot be processed under expanded authority if the award project period end date has expired.

No Cost Extension Request Instructions

This document describes the procedures that must be followed when a grantee determines that their activities will not be completed during the approved project timeline. A no-cost extension requires prior approval from the Director of Grants Management.

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To ensure timely processing of a revised award action and orderly accomplishment of activities, a no-cost extension should be requested at least 45 days prior to the end of the project period.

Grant Details*

Grantee Name	(No response)
Award Program	(No response)
Grant ID	(No response)
Project Title	(No response)
Approved Start Date	(No response)
Approved End Date	(No response)
Total FFAR Funds	(No response)
FFAR Funds Expended To Date	(No response)
FFAR Funds Balance	(No response)
Total Matching Funds	(No response)
Matching Funds Expended To Date	(No response)
Matching Funds Balance	(No response)

Principal Investigator Information*

Prefix	(No response)
First Name	(No response)
Last Name	(No response)
Title	(No response)
Telephone	(No response)
Email	(No response)

Authorized Organization Representative Information*

Prefix	(No response)
First Name	(No response)
Last Name	(No response)
Title	(No response)
Telephone	(No response)
Email	(No response)

New Proposed Project End Date*

Grantee can request a maximum of up to six additional months from the original project end date.

New Project End Date	(No response)
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NCE Justification*

Provide reasons why the project could not be completed in the original period of performance.

(No response)

NCE Activities*

Provide a description of the activities that will be completed during the no cost extension period, if approved.

(No response)

NCE Timeline*

(No response)

FFAR Notification

To ensure timely processing of a revised award action and orderly accomplishment of activities, a no-cost extension should be requested at least 45 days prior to the end of the project period by sending a request on official grantee letterhead that includes the following:

- Date
- Principal Investigator or Project Director name, Project Title, and Grant ID
- Point of contact – name, phone number, and email address
- Amount of additional time requested
- Reason(s) project could not be completed
- Description of the activities that will be completed during the proposed extension
- Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR
- For late requests, a justification for missing the deadline
- Explain the effect a denial of the request will have on the project
- Two signatures – Authorized Signing Official and Project Director/Principal Investigator

NO COST EXTENSION REQUEST

Date:*

(No response)

Principal Investigator:*

(No response)

Project Title:*

(No response)

Grant ID:*

(No response)

PI Contact Information:*

Name	(No response)
Phone Number	(No response)
Email Address	(No response)
Street Address	(No response)
City	(No response)
State	(No response)
Zip Code	(No response)

Amount of additional time requested:*

(No response)

Reason(s) project could not be completed:*

(No response)

Description of the activities that will be completed during the proposed extension:*

(No response)

Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR*:

(No response)

For late requests, a justification for missing the deadline:

(No response)

Explain the effect a denial of the request will have on the project*:

(No response)

Authorized Organization Representative*:



Date Signed*:

(No response)

Principal Investigator*:



Date Signed*:

(No response)

When completing tasks on SM Apply, please note that only the owner of the application (i.e., the user whose name is marked 'owner' in parentheses on the menu on the left of the screen when you open your award) can hit 'Submit' on finished tasks. If you are not the owner, the 'Submit' button will not appear. FFAR does not receive notification of completed tasks without a submission.

Optional - Additional Material Upload

Incomplete

Review Round Final: Scientific Progress Report Review Final for: John Reich

Completed - Apr 10 2019

Score: 0/0

Review Task Form

Scientific Report Review

Award Information*

Grantee Name	The Board of Trustees of the University of Illinois
PI Name	Dr. Amy Marshall-Colon
Grant ID	0000000012
Project Title	Crops in Silico 1.0
Reporting Period	06/06/2017 - 12/31/2018
Award Program	Direct Fund Projects

Project Summary*

Briefly summarize the key goals or questions that the project tried to accomplish or answer

The aim of the Crops in silico 1.0 project was to lay the foundation for generating virtual plant models that accurately capture whole system dynamics in response to in silico environmental and genetic perturbations. A goal of this research project was to identify a feasible approach to link models across biological scales, specifically from genes to phenotypes.

The key goal was to build a computational framework that allows asynchronous communication between models without altering their original code or timesteps. This framework will allow the exchange of inputs and outputs between models, mimicking the flow of biological information across scales.

Did the PI accomplish the approved Goals and Objectives set for this project?*

If the PI met SOME of the Goals and Objectives, select both choices. If the PI met ALL the Goals and Objectives, select 'Yes.' If the PI met NONE of the Goals and Objectives, select 'No.'

Responses Selected:

Yes

Please explain what was accomplished during this project*

The framework to successfully build a multiscale model of the soybean photosynthetic response to elevated CO₂ concentration [CO₂] was created. The vetted model was used to perform in silico genetic perturbations and simulate the anticipated outcomes on photosynthetic efficiency. After dozens of simulations, they identified three top transcription factor gene candidates that will be tested experimentally for their regulatory role in controlling photosynthetic efficiency. The computational framework was also used to link whole plant models of roots and shoots to form the core structure of virtual soybean and corn plants. These linked models were subjected to advanced visualization approaches so that in silico observations can be made on an entire stand of plants. The use of 3-dimensional architecture was shown to improve model simulations of light capture throughout a soybean canopy, providing new insights about plant competition for light resources. Multiple presentations and at least 5 scientific publications

What is the SPD's overall evaluation of what the project was able to accomplish?*

Please provide a brief statement on the Return on Investment for FFAR and the public for funding this project.

They PI's achieved all of their goals and this project was able to accomplish what they set out to accomplish. They were awarded additional funding for the next phase of this project. This is an open source project and will be utilized by the community. It can be vastly impactful in being able to design the appropriate crops in a shorter time frame then possible without the project.

Final Scientific Progress Report Review Rating*

Satisfactory

No equipment records to report.

No metadata records to report.